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Submitter Information

Email: [redacted]
Organization: Hitachi Ltd.

General Comment

See attached file(s)

Attachments

HITACHI RFI Response Development of an Artificial Intelligence Action Plan

March 15, 2025

Mr. Kirk Dohne
Acting Director
Networking and Information Technology Research and Development
National Coordination Office
National Science Foundation
2415 Eisenhower Avenue
Alexandria, Virginia 22314

RE: ***Request for Information on the Development of an Artificial Intelligence (AI) Action Plan***

Dear Mr. Dohne:

The following comments are submitted by Hitachi Group companies (Hitachi) doing business in the United States in response to the February 6th Request for Information on the Development of an Artificial Intelligence (AI) Action Plan.

This document is approved for public dissemination and contains no business-proprietary or confidential information. Document contents may be reused by the government in developing the AI Action Plan and associated documents without attribution.

Hitachi, founded in 1910 and headquartered in Tokyo, Japan, drives Social Innovation Business, creating a sustainable society with data and technology. We solve customers' and society's challenges with Lumada solutions leveraging information technology (IT), operational technology (OT), and products under the business structure of Digital Systems & Services, Green Energy & Mobility, Connective Industries and Automotive Systems. Driven by green, digital, and innovation, we aim for growth through collaboration with our customers. The company's consolidated revenues for FY23 (ended March 31, 2024) totaled $67.1B, with 573 consolidated subsidiaries and approximately 290,000 employees worldwide.

Since establishing a regional subsidiary in the United States in 1959, Hitachi has been a committed American partner. For over thirty years, it has invested heavily in research and development (R&D) in the U.S., and this continued reinvestment has resulted in twenty-four major R&D centers that support high-skilled jobs in manufacturing and technology. Dedicated to delivering the technologies of tomorrow, Hitachi located our digital headquarters in Santa Clara, Calif. to explore applications in ML, AI, Internet of Things (IoT) devices, data analytics, and autonomous vehicles among other advanced technologies. Hitachi is also proud of its human capital investment with more than 17,000 employees across fifty-four companies in the U.S. North America is Hitachi, Ltd.'s second largest market, following only the Japanese market, with $10.9 billion in revenue in FY23.

Hitachi's AI Use Cases

Hitachi has long been a leader in the development and implementation of digital solutions across multiple sectors including manufacturing, energy, and transportation. Within this sphere, the company has historically invested in innovative research related to artificial intelligence (AI). Even before the recent discussions surrounding the technology, Hitachi has recognized the potential for AI—especially in the industrial and generative spaces—as a tool for productivity, efficiency, and automation.

Some of Hitachi's AI use cases are as follows:

- **Predictive maintenance of fleet companies and mass transit:** To address maintenance challenges for fleet companies, Hitachi and Penske applied AI and machine learning (ML) to develop the Guided Repair solution. Guided Repair uses data optimization technology to predict failures before they happen and generate highly accurate diagnoses and recommendations, giving technicians a more efficient and quicker way to solve problems. Hitachi Group company Hitachi Rail uses a similar technology, called Perpetuum, to detect faults while the trains are running remotely and continuously. This data is then used to predict and recognize potential issues.
- **Improving productivity and worker safety on factory floors:** Hitachi's 3D LiDAR (Light Detection and Ranging) technology allows manufacturers to prioritize worker safety while enhancing productivity. Similar to radar and sonar, LiDAR helps factory workers by providing real-time insight that shows bottlenecks and high-risk factors to their safety, operations, and quality without compromising on privacy.
- **An AI tool engineered for the energy industry:** Hitachi Energy recently introduced the Nostradamus AI energy forecasting solution for energy portfolio managers. Purpose-built for the energy industry, Nostradamus AI is a modern AI engine that can generate forecasts estimated to be over 20 percent more accurate than some industry targets. The solution optimizes energy investments, trading strategies, and revenue opportunities, streamlines operational efficiencies and resource planning, and ensures transparency for regulatory compliance.

Hitachi's Recommendations

Effectively harnessing AI is a path for business acceleration and a catalyst for global growth. As our presence in the United States has significantly grown in the last few decades, we support the administration's endeavor to create policy that fosters competitiveness and innovation.

Hitachi appreciates the opportunity to engage with the National Science Foundation and the White House Office of Science and Technology Policy and offer suggestions for the development of an AI Action Plan.

Testbeds & Open-Source Models

Hitachi recommends and has previously advocated for the federal government to enable national AI testbeds. These testbeds can then be freely accessed by any company in the U.S. for use in experiments with AI systems, ensuring consistency of data. This encourages a fair playing field and empowers companies, no matter their size or funding stream, to engage in the same environment as their counterparts in the industry, thus opening the door for competition and giving both emerging and veteran developers an environment to thrive. It has historically been observed that smaller companies, especially startups, find that they cannot compete with larger, more well-known corporations due to a lack of resources, funding, and employee bandwidth. If the federal government were to provide testbeds, however, this would alleviate those burdens to some degree and open the door for increased innovation across several industries, no matter how niche.

In the same vein as the above, Hitachi recommends that the federal government proliferate and encourage the availability of open-source models that all U.S. companies can easily access. Open-source models, like nationally vetted and available testbeds, promote transparency, accessibility, and collaboration, which in turn allows AI to flourish across sectors efficiently and quickly. Public-private partnerships between the federal government and the private sector can also facilitate protected knowledge sharing as well as ensuring AI models are deployed in the safest and most realistic manner possible. Hitachi recognizes the need for closed-door models when appropriate—for example, in safeguarding sensitive business data, or when it comes to national security concerns—but, as practitioners, we understand firsthand the need for open-source models to create an atmosphere of free and fair development.

Knowledge Sharing Across the Federal Government

While public-private partnerships are imperative to the deployment of AI systems, as mentioned above, Hitachi also recommends open doors within the federal government and clear communication across agencies when it comes to AI research and development. The speed of change and innovation associated with AI and the potential of its impact requires collaboration, coordination, and consistent communication among all entities in the federal government. An AI solution created in one agency may apply to a need in a different agency. By openly and regularly sharing and discussing solutions across agencies, the federal government can maximize its resources rather than duplicate efforts by creating the same system via different vendors. It may also capitalize on existing innovations, spur innovative ideas, and limit or eliminate development redundancies.

It is also recommended that agencies continue publishing and/or improve publication of their AI use cases, therefore allowing other agencies and the public to examine the model from an outside perspective and potentially build upon it. This would not only further foster innovation and collaboration, but also promote secure AI, as agencies would be required to create explainable, ethical, and efficient models.

AI as a Strategic Asset and Streamlining AI Approvals

Hitachi strongly recommends that the federal government strengthen federal funding and programs that support AI R&D in critical industries—such as defense, energy, and healthcare—to drive social innovation. This concept of social innovation lies at the heart of Hitachi's core principles, and we firmly believe in advancing digital solutions (including AI) to create resilient communities that improve the lives of Americans and bring the country to an even greater standing among its global peers.

Once again, we underscore the need for a fair and competitive environment, which is supported by the availability and promotion of open-source models and national testbeds, to establish the foundation on which critical industries can build strong and secure AI systems.

In line with the above, Hitachi endorses creating a system of fast-track approval for pathways for AI in healthcare and national security industries, as well as autonomous systems, in addition to other domains where there is a significant advantage of AI automation such as agriculture, energy, education, manufacturing, disease control, skills development and upskilling. Hitachi also recommends easing the FedRAMP process for AI solutions as they are one of the most dynamic and rapidly evolving.

Build a Skilled AI Workforce

At the center of every thriving industry is a skilled and supported workforce. To that end, Hitachi advocates for expanding AI-centric STEM and reskilling programs, investing in AI education and workforce development to enhance digital dexterity regardless of prior expertise or industry experience. While interest in AI may generally be high among employable Americans, the federal government should provide incentives and resources that would allow them to engage with the technology and learn more about its uses and benefits beyond casual use in social media or large language models (LLMs). Immigration reforms targeting the recruitment of high-skilled AI talent should also be a focus to foster a global innovation ecosystem.

Enhance AI Security

All emerging technologies carry some degree of risk. As a company that places immense value on co-creating with its clients, Hitachi understands the need for security in the AI space—not only when it comes to the models and systems themselves, but along the entire supply chain and after deployment as well. We therefore recommend that the federal government prioritize enhancing AI security through export controls and supply chain protection, as well as invest in AI-powered cybersecurity solutions to counter adversarial AI threats, ensuring trust and security.

Safeguarding Intellectual Property (IP) is also crucial in this area. Intellectual Property (IP) leakage is a potential threat that, depending on the system and its use, could pose a national security risk. This is especially true if companies are using third-party vendors to conduct audits which could create IP and security concerns. The AI Action Plan should at minimum address the need to establish guardrails to prevent IP leakage, address privacy and security concerns, and then work with civil society and academia to assist in creating testbeds for third-party verification to ensure AI systems work as intended and avoid compromising IP.

Conclusion

Hitachi appreciates the opportunity to provide information to the NITRD and NSF on the development of a federal AI action plan. We look forward to further iterations of the plan and continuing to work with the Trump-Vance administration and the National Science Foundation to provide meaningful feedback as needed in this space.

Sincerely,

[redacted]

Shashank Samant
Executive Chairman
Hitachi America